



DSA Day 2 Preparation

Learning Data Structures and Algorithms (DSA) using Python and Java



Day 2 focuses on control flow, conditions, and decision making in Python and Java.



1 Topics Covered

- What is Control Flow?
- What is a Condition?
- Why Conditions are Important
- Comparison Operators
- If Statement
- If-Else Statement
- If-Elif-Else in Python
- Else If in Java
- Logical Operators: AND, OR, NOT
- Condition-Based Practice Problems

2 What is Control Flow?



Control flow means the order in which a program runs.

Programs follow a sequence, but sometimes they need to make decisions.

3 What is a Condition?



A condition is an expression that returns either True or False.

Example:

```
age >= 18
```

4 Comparison Operators

Operator	Meaning	Example
==	equal to	a == b
!=	not equal to	a != b
>	greater than	a > b
<	less than	a < b
>=	greater than or equal to	a >= b
<=	less than or equal to	a <= b

5 If Statement

Python

```
age = 20
if age >= 18:
    print("You are eligible
    to vote.")
```

Java

```
int age = 20;
if (age >= 18) {
    System.out.println(
        "You are eligible
        to vote.");
}
```

6 If-Else Statement

Python

```
marks = 65
if marks >= 50:
    print("Pass")
else:
    print("Fail")
```

Java

```
int marks = 65;
if (marks >= 50) {
    System.out.println("Pass");
} else {
    System.out.println("Fail");
}
```

7 Multiple Conditions

Python uses **elif** and Java uses **else if** for multiple branches.

Python

```
marks = 82
if marks >= 90:
    grade = 'A'
elif marks >= 75:
    grade = 'B'
elif marks >= 60:
    grade = 'C'
else:
    grade = 'F'
```

Java

```
int marks = 82;
char grade;
if (marks >= 90) {
    grade = 'A';
} else if (marks >= 75) {
    grade = 'B';
} else if (marks >= 60) {
    grade = 'C';
} else {
    grade = 'F';
}
```

9 Logical Operators

Operator	Python	Java
AND	and	&&
OR	or	
NOT	not	!

- AND:** both conditions must be true.
- OR:** at least one condition must be true.
- NOT:** reverses the result (True → False, False → True).

10 Why Conditions Matter in DSA



- Compare values
- Check even/odd
- Find maximum
- Make decisions inside algorithms
- Control program behavior based on input

11 Practice Problems

- Check Even or Odd
- Check Pass or Fail
- Voting Eligibility
- Find Largest of Two Numbers
- Positive, Negative, or Zero

12 Day 2 Summary

- ✓ Conditions help programs make decisions
- ✓ A condition returns true or false
- ✓ **if** runs when a condition is true
- ✓ **else** runs when it is false
- ✓ logical operators combine conditions

13 Learning Journey



Day 1

Programming Basics, Complexity, Constraints



Day 2

Conditions and Decision Making



Day 3

Loops and Loop-Based Problems



Day-2

05/05/2026.

→ Today we will discuss about control flow, especially
Conditions.

→ Loops.

These two concepts are very important because almost every DSA problem uses them.

→ In Day 1, we discussed that programming means giving instruction to a computer.

→ But real problems are not always simple.

Sometimes a program must decide:

Eg: Is the number even (or) odd?

Is the student passed (or) failed?

Is the user eligible (or) not?

Some times a program must repeat work:

Eg: Print numbers from 1 to n.

Check every element in an array

Calculate sum of many numbers.

For decisions, we use Conditions.

For repetition, we use loops.

① What is Control Flow?

→ Control flow means the order in which instructions are executed in a program.

↳ Normally, code runs from top to bottom.

Example:

Step 1: Start

Step 2: Take input.

Step 3: Process data.

Step 4: Print Output.

Step 5: End.

But sometimes we do not want every line to run.

Example:

IF marks are greater than 35 or equal to 35,

Print Pass. Otherwise, print fail.

Here, the output depends on the condition. So Control flow helps the program choose which path to follow.

② What is Condition?

→ A Condition is a statement that gives only one of two answers.

True (or) False

Example:

$\text{age} \geq 18$

→ If $\text{age} = 20$

$20 \geq 18 \rightarrow \text{True}$

→ If $\text{age} = 15$

$15 \geq 18 \rightarrow \text{False}$

→ So a Condition is like a question asked by the program

Example:

Is age greater than (or) equal to 18?

Is marks greater than (or) equal to 35?

Is number divisible by 2?

→ The answer is either true (or) false.

③ Why Conditions are Needed?

↳ Without conditions, a program cannot make decisions.

Example:

A voting program should not print the same result for everyone.

If age is 20:

↳ Eligible to vote.

If age is 15:

↳ Not eligible to vote.

→ So we need conditions to make the program behave differently for different inputs.

④ Comparison Operators

→ Comparison operators are used to compare two values.

Operators	Meaning	Example	Result
$==$	Equal to	$10 == 10$	True
$!=$	Not equal to	$10 != 5$	True
$>$	Greater than	$10 > 5$	True
$<$	Less than	$10 < 5$	False

$>=$ Greater than (or) Equal to $10 >= 10$ True

$<=$ Less than (or) Equal to $5 <= 10$ True

Very important difference:

→ $=$ means assign a value, Eg: $\text{age} = 20$ (store 20 in variable age)
 $==$ means Compare two values, Eg: $\text{age} == 20$ (whether age is equal to 20)

⑤ If statement

→ The if statement is used to run code only when a condition is true.

→ Think like this: If condition is true, then do this work.

Example:

→ If age is 18 (or) above, Print Eligible.

Python:

```
age = 20
```

```
if age >= 18:
```

```
    print("Eligible to vote")
```


Java

class Main {

public static void main (String[] args) {

int age = 20;

if (age >= 18) {

System.out.println("Eligible to vote");

}

}

Explanation:

Age = 20

Condition : age >= 18

20 >= 18 is true

So the print statement runs

Output: Eligible to vote.

⑥ If-Else Statement

→ The if-else statement is used when there are two possible results

→ Think like this:

If condition is true, do this.

Otherwise, do that

Example:

→ If marks are 35 or more, print Pass.

Otherwise, print fail.

Python
marks = 30

if marks $\geq$ 35:

 print("Pass")

else:

 print("Fail").

Explanation:

marks = 30

Condition: marks $\geq$ 35

30 $\geq$ 35 is false

so else block runs → output: fail

Java:

Class Main {

 public static void main (String [] args) {

 int marks = 30;

 if (marks $\geq$ 35) {

 System.out.println("Pass");

 } else {

 System.out.println("fail");

⑦ Logical Operators:

→ Some times one condition is not enough.

Example:

A student is allowed for exam only if:

→ marks ≥ 35 and attendance ≥ 75

Here two conditions are used together.

Logical operators help combine conditions.

Meaning

AND

OR

NOT

Python

and

or

not

Java

&&

||

!

Use:

Both conditions must be true

at least one condition should be true

Reverse the result

if (28 < 30) {

print("pass")

print("fail")

if (28 < 30 && 75 < 75) {

print("pass")

print("fail")

if (28 < 30 || 75 < 75) {

print("pass")

print("fail")